I-ACO algorithm explained

$$\frac{|\text{tour_static}_{static} - \text{tour_init}_{a}}{\text{tour_init}_{aver}} \qquad \text{our_init}_{aver} = \frac{\sum_{i=0}^{n} \text{tour_in}}{n}$$

$$\frac{|\text{tour_aco}_{aver} - \text{tour_init}_{aver}}{\text{tour_init}_{aver}} \qquad \text{ır_static}_{aver} = \frac{\sum_{i=0}^{n} \text{tour_st}}{n}$$

$$\frac{|\text{tour_opt}_{aver} - \text{tour_init}_{aver}}{\text{tour_init}_{aver}} \qquad \text{our_aco}_{aver} = \frac{\sum_{i=0}^{n} \text{tour_ac}}{n}$$

$$\frac{|\text{tour_aco}_{aver} - \text{tour_opt}_{aver}}{\text{tour_aco}_{aver}} \qquad \text{our_opt}_{aver} = \frac{\sum_{i=0}^{n} \text{tour_o}}{n}$$

i dont undderstand these three formulas niw give me exampel toundertand tgese abd tell ne what is the use of it

Great — I will explain these formulas in **very simple words**, and then I'll give you a **small numeric example** so you understand exactly what they mean and why they are used.

✅ FIRST: What these formulas measure

These formulas measure **how much improvement** or **how much congestion reduction** each algorithm provides.

To do this, they compare:

- **Initial path cost** (before any optimization)
- **Static ACO path cost**
- **Dynamic ACO path cost (improved ACO)**
- **Optimized ACO path cost (PSO + ACO)**

They do this by calculating **average path cost** for n test cases.

✅ SECOND: Meaning of Each Average

✔ **tour_init_aver = average cost of initial paths**

→ Before using ACO or PSO.

✔ **tour_static_aver = average cost using static path (distance-based ACO)**

✔ **tour_aco_aver = average cost using improved dynamic ACO (with road factor R)**

✔️ tour_opt_aver = average cost using PSO-optimized ACO

These average values are computed over **n experiments**, for example n = 500 random start/end pairs.

📌 FORMULAS for averages

Each average is simply:

$$tour_something, aver = \frac{\sum tour_something(i)}{n}$$

This means:

➡️ Add the cost for all n paths
➡️ Divide by n
➡️ You get the **average path cost**

📌 Formulas for Congestion Reduction (CR)

1. CR_static

$$CR_{static} = \frac{|tour_static, static - tour_init, aver|}{tour_init, aver}$$

Measures **how much static ACO improved** compared to initial.

---

2. CR_aco

$$CR_{aco} = \frac{|tour_aco, aver - tour_init, aver|}{tour_init, aver}$$

Measures **how much dynamic ACO improved** compared to initial.

---

3. CR_opt

$$CR_{opt} = \frac{|tour_opt, aver - tour_init, aver|}{tour_init, aver}$$

Measures **how much PSO-optimized ACO improved** compared to initial.

---

4. **CR_promote**

$$CR_{promote} = \frac{|tour_aco, aver - tour_opt, aver|}{tour_aco, aver}$$

Measures **how much improvement PSO gives** over the improved ACO.

⭐ WHY DO WE NEED THESE? (Easy Explanation)

We need these formulas to:

- ✔ Compare different algorithms
- ✔ Measure how much better a new method is
- ✔ Calculate congestion reduction
- ✔ Compare ACO vs improved ACO vs PSO+ACO
- ✔ Show improvements in experiments in a single number

✅ NOW: A VERY SIMPLE NUMERIC EXAMPLE

Let's say we test **n = 5 paths**.

Initial path costs

tour_init = [120, 130, 110, 150, 140]

$$tour_init, aver = (120 + 130 + 110 + 150 + 140)/5 = 130$$

Static ACO path costs

tour_static = [100, 110, 90, 120, 115]

$$tour_static, aver = (100 + 110 + 90 + 120 + 115)/5 = 107$$

Dynamic ACO path costs

tour_aco = [80, 90, 85, 95, 92]

$$tour_aco, aver = (80 + 90 + 85 + 95 + 92)/5 = 88.4$$

PSO-optimized ACO path costs

tour_opt = [78, 85, 82, 88, 90]

$$tour_opt, aver = (78 + 85 + 82 + 88 + 90)/5 = 84.6$$

⭐ Now calculate the CR values

1. CR_static

$$CR_{static} = \frac{|107 - 130|}{130} = \frac{23}{130} = 17.69\%$$

Static ACO improves congestion by **17.69%**

---

2. CR_aco

$$CR_{aco} = \frac{|88.4 - 130|}{130} = 41.6/130 = 32\%$$

Dynamic ACO improves congestion by **32%**

---

3. CR_opt

$$CR_{opt} = \frac{|84.6 - 130|}{130} = 45.4/130 = 34.9\%$$

PSO-optimized ACO improves congestion by **34.9%**

---

4. CR_promote

$$CR_{promote} = \frac{|88.4 - 84.6|}{88.4} = 3.8/88.4 = 4.29\%$$

PSO improves dynamic ACO by **4.29%**

⭐ FINAL SUPER-EASY SUMMARY

- **tour_init_aver** = average initial cost

- **tour_static_aver** = average cost using distance-based ACO
- **tour_aco_aver** = average cost using improved ACO
- **tour_opt_aver** = average cost using PSO + ACO

The CR formulas measure **how much each method reduces congestion** compared to initial or compared to each other.

If you want, I can draw a diagram showing how these metrics compare.

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PDF

i am understand ing that instead of distance road condition factor nw tell me what is the use of road condtion fcrot and all the formulas that are menthioned in the paper and also tell me that how they are applied with the algorithm also and tellme these in simple and easy and in precise .

Nice, you've basically got the key idea already: **we replace "distance" with a smarter cost called the road-condition factor R.**
Now let's tie *all* the formulas around R together, in simple but precise way.

1️⃣ What is the road-condition factor R used for?

Instead of saying:

> "shortest distance = best path"

the paper says:

> "**lowest R = best path**"

where **R = road condition factor** that includes:

- road length d_i
- number of lanes l_i

- cars already on the road
- incoming & outgoing traffic (congestion)

So:

- If a road is **short but very crowded**, its **R is large** → it looks *expensive*.
- If a road is **longer but free**, its **R is smaller** → it can be chosen as a better route.

They use **R everywhere that classic ACO / PSO would use distance**.

2️⃣ Formulas for the road-condition factor

(a) Congestion coefficient $C_i(t)$ – *how crowded is road i now?*

$$C_i(t) = \begin{cases} \frac{(f_i(t-1)+f_i^{in}(t)-f_i^{out}(t))\cdot L}{l_i \cdot d_i}, & \text{if } f_i(t-1)+f_i^{in}(t) > f_i^{out}(t) \\ 0, & \text{otherwise} \end{cases}$$

Meaning:

- $f_i(t-1)$: cars already on road i at previous time
- $f_i^{in}(t)$: cars entering road i now
- $f_i^{out}(t)$: cars leaving road i now
- L: average car length
- l_i: number of lanes
- d_i: physical length of the road

Roughly this is "**cars per available space**".
If outgoing cars ≥ incoming + existing → no build-up → set congestion to 0.

(b) Road-condition factor $R_i(t)$

$$R_i(t) = d_i \cdot \big(1 + C_i(t)\big)$$

If no congestion ($C_i = 0$) → $R_i = d_i$ (just distance).
If congestion exists → $C_i > 0$ → $R_i > d_i$ (road becomes more "expensive").

Alternate expanded form (just plugging (a) inside):

$$R_i(t) = d_i \left(1 + \frac{(f_i(t-1) + f_i^{in}(t) - f_i^{out}(t)) \cdot L}{l_i \cdot d_i}\right)$$

(c) Change of road condition ΔR_i

$$\Delta R_i = R_i(t+1) - R_i(t)$$

This tells **how much a road got worse or better** between two time steps:

- big positive ΔR_i → sudden new congestion
- small ΔR_i → stable traffic

This is used to decide **when to re-plan the route**.

3️⃣ How R is used *inside the algorithms*

🔹 In PSO (parameter optimisation)

In the PSO part:

- They plug **R instead of distance** into the fitness function.
- PSO searches for the best ACO parameters α, β, ρ that give **small total R** on routes.

So PSO learns:

> "What values of α, β, ρ make the ants find paths with low congestion (low total R)?"

Result: ACO gets **good parameters** for congested traffic.

🔹 In ACO (path search)

Everywhere normal ACO uses distance d_{ij}, they use **road condition R**:

1. **Heuristic information**

Classical: $\eta_{ij} = 1/d_{ij}$
Here: $\eta_{ij} = 1/R_{ij}$

So roads with **low R (free, wide)** get **high heuristic value** and are more attractive.

2. **Path length L_k**

For an ant's tour:

- Classical: $L_k = \sum d_{ij}$
- Here: $L_k = \sum R_{ij}$

So pheromone update rewards routes with **small total R** (less congestion), not just short distance.

3. **Dynamic re-planning**

At each time step they compute $R_i(t)$ and ΔR_i.

If:

- $R_i(t) > R_{max}$ (road too congested) **or**
- $\Delta R_i > \Delta R_{max}$ (road suddenly worsened),

then:

> **Run ACO again** to find a new route (newtour) using the latest R values.

Thus the route adapts to live traffic.

4️⃣ How R appears in the experiment formulas

For each experiment they define:

- **tour_init(i)** – sum of R_i along the **initial** route
- **tour_static(i)** – sum of R_i along **distance-based ACO** route
- **tour_aco(i)** – sum of R_i along **improved ACO** route
- **tour_opt(i)** – sum of R_i along **PSO+improved ACO** route

Then they average over n tests:

$$tour_aco, aver = \frac{1}{n}\sum_{i=1}^{n} tour_aco(i)$$

(similarly for the others)

Finally they compute congestion rates, e.g.:

$$CR_{aco} = \frac{|tour_aco, aver - tour_init, aver|}{tour_init, aver}$$

So **all these CR values are based on sums of R**, not sums of distance.
This allows them to say things like:

> "Our method reduces congestion rate by 9.73–13.63%."

because they are measuring **actual road condition cost**, not just kilometers.

5️⃣ Tiny numeric example to fix the idea

Suppose two roads from A to B:

- Road 1: $d_1 = 2$ km, very crowded → $C_1 = 1.0$
  → $R_1 = 2 \times (1 + 1) = 4$
- Road 2: $d_2 = 3$ km, almost free → $C_2 = 0.1$
  → $R_2 = 3 \times 1.1 = 3.3$

Distance-based ACO: prefers Road 1 (2 km < 3 km).
R-based ACO: prefers Road 2 (3.3 < 4 → lower cost, less congestion).

That's exactly why R is introduced.

🧠 One-sentence takeaway

Road condition factor R turns "shortest distance" into "least congested, most realistic route", and it is used everywhere in the algorithm (heuristic, path cost, dynamic re-planning, and evaluation) to make ACO work in real city traffic instead of on a clean static map.

giveme small mathbased on it .that cover all of theabove aspects so that i can understand the work properly

Nice, you've got the big picture. Let's do **one small numeric example** that touches **all the pieces**:

- road-condition factor R
- congestion change ΔR
- how R is used in ACO (heuristic + tour length)
- how this leads to better paths and the CR formula

1️⃣ Tiny road network

3 intersections:

- 0 = start
- 1, 2 = customers

Roads (distance in km):

- 0–1: $d_{01} = 2$
- 0–2: $d_{02} = 3$
- 1–2: $d_{12} = 2$

There are 2 possible routes from 0 to visit both 1 and 2 and return:

- **Route A**: 0 → 1 → 2 → 0
- **Route B**: 0 → 2 → 1 → 0

Traditional ACO uses only **distance**, so:

- Route A distance = $2 + 2 + 3 = 7$ km
- Route B distance = $3 + 2 + 2 = 7$ km

Distances are the same, so distance alone cannot tell which is better in congestion.

2️⃣ Road-condition factor R

Assume congestion coefficients (already computed from flows using the paper's formulas) are:

- Road 0–1: **very congested** → $C_{01} = 1.0$
- Road 0–2: **light congestion** → $C_{02} = 0.1$
- Road 1–2 and 2–1: **medium** → $C_{12} = C_{21} = 0.2$

Road-condition factor:

$$R_{ij} = d_{ij}(1 + C_{ij})$$

Compute:

- $R_{01} = 2(1 + 1.0) = 4.0$
- $R_{02} = 3(1 + 0.1) = 3.3$
- $R_{12} = R_{21} = 2(1 + 0.2) = 2.4$

👉 Even though 0–1 is **shorter** (2 km), it is **more expensive** in terms of R (4.0) than 0–2 (3.3) because of congestion.

3️⃣ How ACO uses R: heuristic & probability

Heuristic:

$$\eta_{ij} = \frac{1}{R_{ij}}$$

So:

- $\eta_{01} = 1/4.0 = 0.25$
- $\eta_{02} = 1/3.3 \approx 0.303$

Let pheromone on both edges initially be the same:

- $\tau_{01} = \tau_{02} = 0.1$

Take $\alpha = 1$, $\beta = 2$.
From node 0 an ant must choose between 1 and 2.

Attractiveness:

$$A_{01} = \tau_{01}\eta_{01}^{\beta} = 0.1 \cdot (0.25)^2 = 0.1 \cdot 0.0625 = 0.00625$$

$$A_{02} = \tau_{02}\eta_{02}^{\beta} \approx 0.1 \cdot (0.303)^2 \approx 0.1 \cdot 0.0918 = 0.00918$$

Probabilities:

$$p(0 \to 1) = \frac{A_{01}}{A_{01} + A_{02}} = \frac{0.00625}{0.00625 + 0.00918} \approx 0.405$$

$$p(0 \to 2) \approx 0.595$$

👉 **Because 0–2 has smaller R (less congestion), the ant is more likely to go 0→2.**
So ACO naturally avoids the congested road.

4️⃣ Route cost using R

Use R instead of distance for tour length:

- **Route A** (0 → 1 → 2 → 0):
  $R_A = R_{01} + R_{12} + R_{20} = 4.0 + 2.4 + 3.3 = 9.7$
- **Route B** (0 → 2 → 1 → 0):
  $R_B = R_{02} + R_{21} + R_{10} = 3.3 + 2.4 + 4.0 = 9.7$

Right now both equal (in this toy example), but the key point is:

- The route that **uses 0–1 first** suffers congestion earlier, which can affect time windows, etc.

- In practice they also consider time, dynamic changes, etc., so small R differences matter.

For our CR example we'll simplify a bit more.

5️⃣ Dynamic change & ΔR

Suppose at the next time step $t+1$:

- Congestion on 0–2 suddenly increases: C_{02} jumps from 0.1 → 0.8
- New $R_{02}(t+1) = 3(1+0.8) = 5.4$

Change:

$$\Delta R_{02} = R_{02}(t+1) - R_{02}(t) = 5.4 - 3.3 = 2.1$$

If the threshold is, say, $\Delta R_{\max} = 2.0$:

- Since $\Delta R_{02} > \Delta R_{\max}$, the algorithm **re-plans the route** using updated R values.

Now 0–2 is more congested than 0–1, so new R-based ACO may switch from route B to route A.
That's how **dynamic path planning** works.

6️⃣ Comparing before & after using CR formula

Let's do a super-simple n = 1 case so you can see the CR formula.

Assume:

- **Initial route** (no congestion considered, distance only) chooses 0–1–2–0
  and we measure its cost with R:
  $tour_init(1) = R_A = 9.7$
- **Improved dynamic ACO route** (using R and re-planning) finds a slightly better path or timing:
  suppose its total cost is
  $tour_aco(1) = 8.5$

Then:

$$tour_init, aver = 9.7, \quad tour_aco, aver = 8.5$$

Congestion reduction rate of improved ACO:

$$CR_{aco} = \frac{|tour_aco, aver - tour_init, aver|}{tour_init, aver} = \frac{|8.5 - 9.7|}{9.7} = \frac{1.2}{9.7} \approx 12.37\%$$

👉 This says: **dynamic ACO reduced the average "congested cost" by about 12% compared to the initial path**.

This is exactly how the paper reports numbers like **"9.73%–13.63% reduction"**:
they compare average R-based tour costs before and after the improved algorithm.

🔚 One-sentence takeaway

- **R** turns each road into a "real traffic cost" (distance + congestion).
- ACO uses R in the heuristic and tour length instead of distance.
- ΔR tells when traffic changes enough to re-plan.
- CR formulas compare average total R before and after, giving the % congestion reduction.

If you'd like, I can now map this tiny example directly onto the exact notation in your paper (tour_init, tour_aco, CR_aco, etc.) so you can use it in your notes.